

Post-Quantum Advanced Encryption Scheme from Code-based Assumptions

Master's Internship Proposal, Télécom Paris

2027

General information

Location: Télécom Paris (Palaiseau)

Team: Cybersecurity and Cryptography [C²] team, INFRES department

Advisors:

- Victor Dyseryn (<https://victordyseryn.github.io>)
- Duong Hieu Phan (<https://www.di.ens.fr/~phan>)

Starting date, duration: from February-March 2027 (flexible), 6 months. The internship may be followed by a PhD on advanced primitives in post-quantum cryptography from code-based assumptions, for which we already have funding.

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Context

Large-scale quantum computers would break the public-key cryptography deployed today, such as RSA and Elliptic Curve Cryptography, whose security relies on problems that Shor's algorithm solves in polynomial time. **Post-quantum cryptography** (PQC) aims to design replacements that resist quantum adversaries. For *basic* primitives (public-key encryption and digital signatures), several families of assumptions are available: lattices, but also **codes** (e.g., HQC encryption scheme [1] or the SDitH signature scheme [2]), isogenies or multivariate systems.

The situation is very different for **advanced primitives**, i.e. constructions that build on these basic components to **offer richer functionalities** such as fine-grained access control or accountability. Here, almost all known post-quantum constructions rely on lattice assumptions. This monoculture is risky: a single cryptanalytic breakthrough on lattices would wipe out the whole landscape of advanced post-quantum cryptography at once. Diversifying the underlying assumptions is therefore a pressing objective, and may also bring different efficiency trade-offs.

This internship focuses on one such advanced encryption primitive: **broadcast encryption**. A broadcast encryption scheme involves N users, each holding a distinct secret key. Using a single public key, anyone can encrypt a message for any subset S of the N users, so that only users in S can decrypt, even if all users outside S collude. The main efficiency goal is to keep ciphertexts short, ideally with a size independent of the number of recipients. Broadcast encryption has many applications in the secure distribution of content to dynamic sets of recipients. Examples include pay-TV, satellite broadcasting, and secure group communication.

Research project

The main objective of this internship is to **design** and **analyze** a post-quantum **broadcast encryption** scheme whose security relies on **code-based assumptions**, that is, on the hardness of decoding random linear codes. Two settings can be considered: the *Hamming metric*, where the reference assumptions are the Syndrome Decoding problem and its close variant, Learning Parity with Noise (LPN); and the *rank metric*, where the reference assumption is the Rank Syndrome Decoding problem. These problems have been studied for decades, and their algebraic structure differs markedly from that of lattices. Constructions based on them would therefore be a genuine alternative to LWE-based schemes.

Starting point. A very recent result [3] established the feasibility of *traitor tracing* from code-based assumptions in the Hamming metric. Traitor tracing is a primitive closely related to broadcast encryption that makes it possible to identify users who leak their decryption keys. A key ingredient of this work is a *kernel sampling* technique. The intern will first study this result in depth. They will also study state-of-the-art broadcast encryption schemes from lattices [4], with the goal of identifying which lattice techniques could have a code-based counterpart.

Main research directions. The core of the internship will be to transpose the kernel sampling technique from the Hamming metric to the **rank metric**. The rank metric has a richer algebraic structure, since error supports are subspaces rather than sets of coordinates. This structure raises new technical challenges, but it may also lead to more compact parameters. Building on this, the intern will investigate:

- whether the technique yields a **broadcast encryption** scheme with short ciphertexts;
- whether it can lead to an **identity-based encryption** (IBE) scheme from code-based assumptions, a long-standing open problem in the field;
- develop **security proofs** and possibly **problem reductions** to prove their hardness;
- optionally develop a proof-of-concept **implementation**.

Required skills

- Linear algebra, finite fields
- Strong knowledge in cryptography
- (preferred) Basic knowledge of post-quantum cryptography (lattice-based or code-based)
- (optional) C++ and/or Python

References

- [1] Gaborit, P., Aguilar-Melchor, C., Aragon, N., Bettaieb, S., Bidoux, L., Blazy, et al. (2025). Hamming Quasi-Cyclic (HQC). *NIST Post-Quantum Standardization, 4th Round*.
- [2] C. Aguilar Melchor, T. Feneuil, N. Gama, S. Gueron, J. Howe, D. Joseph, A. Joux, E. Persichetti, T. H. Randrianarisoa, M. Rivain, D. Yue (2023). SDitH. First-round submission to NIST call for additional post-quantum signature schemes (<https://csrc.nist.gov/projects/pqc-dig-sig>)
- [3] Thomas Debris-Alazard, Victor Dörsner, Duong Hieu Phan (2026). Hardness of Problems with Hints in Code-Based Cryptography and Applications to Traitor Tracing. To appear at *ASIACRYPT 2026*.
- [4] Shweta Agrawal, Shota Yamada (2020). Optimal Broadcast Encryption from Pairings and LWE. In *Advances in Cryptology – EUROCRYPT 2020*, LNCS, Springer.